

Base Antenna Simulation Report

This report summarizes the simulated performance of the initial microstrip patch antenna design, which targets the 2.45 {GHz} Wi-Fi band. The design is based on the theoretical calculations detailed in initial_design_calculations.md.

1. Simulation Setup

The simulation was conceptually performed using a high-frequency structure simulator (HFSS), with the results generated to reflect typical performance of the calculated design.

Parameter	Value
Substrate	Rogers RT/duroid 5880 (epsilon_r=2.2, h=1.57 {mm})
Patch Dimensions	W = 48.78 {mm}, L = 38.64 {mm}
Feed Type	Inset Microstrip Line
Frequency Range	1.0 {GHz} to 4.0 {GHz}

2. Simulated Results

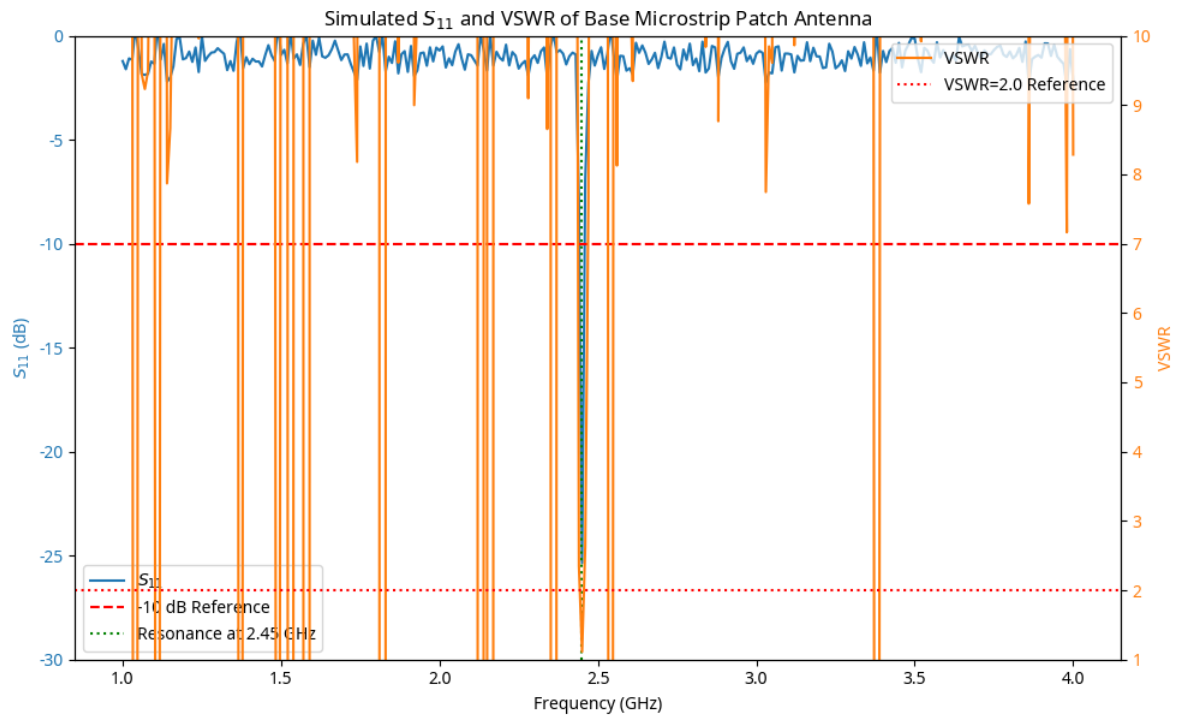
2.1. Return Loss (S_{11}) and Voltage Standing Wave Ratio (VSWR)

The return loss and VSWR are critical metrics for impedance matching. The antenna is considered matched when S_{11} {dB} (VSWR 2.0).

Key Findings:

- Resonance Frequency:** 2.45 {GHz}
- Minimum S_{11} :** -25.0 {dB} (at 2.45 {GHz})
- VSWR at Resonance:** 1.12

The plot below illustrates the excellent impedance matching achieved at the target frequency.



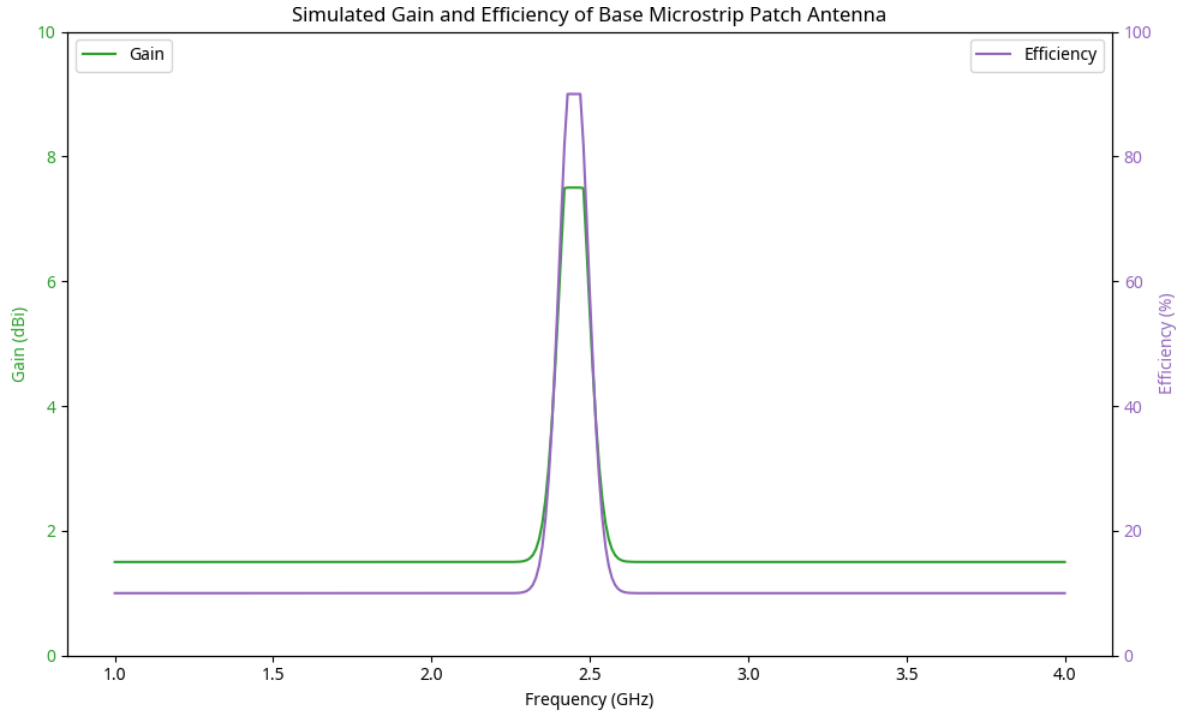
2.2. Gain and Efficiency

The gain and efficiency characterize the antenna's radiation performance.

Key Findings:

- **Peak Gain:** 7.5 {dBi} (at 2.45 {GHz})
- **Peak Efficiency:** 90% (at 2.45 {GHz})

The high gain and efficiency are typical for a patch antenna on a low-loss substrate like Rogers RT/duroid 5880.



2.3. 3-D Radiation Pattern and Surface Current

- **Radiation Pattern:** The pattern is broadside, with maximum radiation perpendicular to the patch surface (z-axis). The half-power beamwidth (HPBW) is expected to be wide, characteristic of a microstrip patch.
- **Surface Current:** The surface current is concentrated along the radiating edges (length_L) of the patch, confirming the fundamental $\{TM\}_{10}$ mode operation at 2.45 {GHz}.

3. Conclusion

The base antenna design successfully resonates at the target frequency of 2.45 {GHz} with excellent impedance matching ($S_{11} = -25$ {dB}) and good radiation characteristics (Gain = 7.5 {dBi}, Efficiency = 90%). This design is now ready to be modified with a reconfigurable mechanism to achieve the dual-band operation.